

Once you
stop learning,
you start dying.

—Albert Einstein

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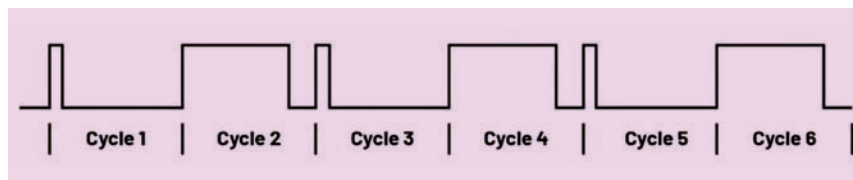


Fig. 3: A switch node voltage—a subharmonic oscillation with a current-mode regulator

be much more easily compensated and thus stabilised. Fig. 2 shows the simple transfer function of the power stage of a typical current-mode regulator.

However, a few disadvantages exist alongside the mentioned advantages. Current-mode regulators cannot immediately make the required current measurements after a switching transition because the noise will couple into the measurement strongly at this time. It takes a few nanoseconds for the noise caused by the switching to subside. This is called the blank-

ing time. It normally results in a somewhat longer minimum on-time specification than for voltage-mode regulators.

Another disadvantage of current-mode regulators is the possibility, in principle, of a subharmonic oscillation. This is shown in Fig. 3. If a duty cycle of greater than 50% is required, a current-mode regulator may alternately execute short and long pulses. In many applications, this is considered instability, which should be avoided.

To solve this, a certain ramp compensation can be added to the

generated current ramp shown in Fig. 1. It can shift the critical duty cycle threshold to well above 50% so that even at higher duty cycles, no subharmonic oscillations occur.

Even these earlier mentioned restrictions, due to the blanking time and the resultant duty cycle limitations, can be circumvented through the IC design. For example, one remedy is to incorporate low-side current sensing where the inductor current is measured during the off time rather than the on time.

All in all, the advantages of current-mode control in switching regulators outweigh the disadvantages for most applications. And through various circuit innovations and modifications, the disadvantages can be bypassed. As a result, most switching regulator ICs today use current-mode control. **EFY**